

Pun-ny Math

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1 Introduction

Here is the key that you will need to use to solve the following problem set. You will solve the given problems in order to answer the questions (that are in words). You will need to rearrange the letters (keep in mind that some letters will repeat) you get while solving the individual problems in order to arrive at your final answer.

A = 0	F = 5	K = 10	P = 15	U = 20	Z = 25
B = 1	G = 6	L = 11	Q = 16	V = 21	Note: All the problems are given in ascending order of the resulting numbers
C = 2	H = 7	M = 12	R = 17	W = 22	
D = 3	I = 8	N = 13	S = 18	X = 23	
E = 4	J = 9	O = 14	T = 19	Y = 24	

2 Problems

1. Question: What do you get when you multiply a famous place in New York City by itself?

• Answer: -----

(a) (Part i's answer) - [(Part b's answer) + (Part e's answer)] =

(b) (Letter used twice): The only number whose principal square root is half the number itself =

(c) $2^4 - 2^3 =$

(d) (f's answer) $- 5 =$

(e) $(5! - 4!) \div 6 =$

(f) $\sqrt{300 - 11} =$

(g) (Letter used twice): $6 + 324 \div 6 - 42 =$

(h) Find the positive difference between 2116 and the year this year's entering freshman class (2021-2022) will graduate high school. Now flip the resulting number's digits (for example, the number 48 would result in 84) =

(i) $(2021 \div 47) - (2020 \div 202) - (2019 \div 673) - (2018 \div 1009) - (2017 \div 2017) - (2016 \div 288) =$

2. Question: What do you call brothers who love math?

• Answer: -----

(a) (g's answer) $- [(d's answer) + (e's answer)] =$

(b) $|1^2 - 2^2| \div 3 =$

(c) $(6 - 5) \times 4 =$

(d) $7 + 8 - 9 =$

(e) $10 - 11 + 12 =$

- (f) $13 + |14 - 15| =$
- (g) $16 - 17 + 18 =$
- (h) $19 + 20 - 21 =$

3. Question: What did the triangle say to the circle?

- Answer: ---'--- - - - - -
- (a) (Letter used twice): (f's answer)-(c's answer) =
- (b) $1 + 2^2 + 3 =$
- (c) First Digit = $|4 - 5|$, Second Digit = $|6 - 7|$
- (d) $8 - 9 + 10 - 11 + 12 - 13 + 14 - 15 + 16 - 17 + 18 =$
- (e) (Letter used twice): $18 - ||19^2 - 20^2| - |21^2 - 22^2| =$
- (f) $|23 - 24| \times (25 + 26)$, flip the resulting number's digits =
- (g) $(2930 - 2728 - 32) \div 10 =$
- (h) (Letter used twice): $31 + 33 - 35 - 11 =$
- (i) $38 \div (36 - 34) =$
- (j) $40 \div (44 - 42) =$
- (k) $|42 - 46|! =$

4. Question: What do you call a political party in favor of agriculture?

- Answer: ---'- - - - -
- (a) (g's answer)-[(b's answer)+(e's answer)] =
- (b) This number is the only even prime number =
- (c) (Letter used twice): This number has a prime factorization of (b's answer)(7) =
- (d) This number is the smallest two digit positive multiple of 5 that has only odd factors =
- (e) (Letter used twice): This is the second largest prime under 20 =
- (f) This number has a prime factorization of (b's answer)(e's answer-c's answer)² =
- (g) (Letter used twice) This is the largest prime under 20 =

5. Question: What do you call a male mathematician who spent all summer on the beach?

- Answer: - - - - -
- (a) (Letter used twice): (e's answer)-[(c's answer)+(d's answer)] =
- (b) The function of $p \heartsuit q$ is defined as $2p + 3q$. If $x \heartsuit 36 = 188$, find $x - 36 =$
- (c) The function of $h \spadesuit k$ is defined as $h^2 - k^2$. If $x \spadesuit 21 = 184$, find $|x| - 19 =$
- (d) (Letter used twice): The function of $t \star u$ is defined as $-3(t + u)$. If $5 \star x = 9$, find $5 - x =$
- (e) (Letter used twice): The function of $y \blacksquare z$ is defined as $2(y - 2)^2$. If $x \blacksquare 20 = 722$ and $y < z$, find $20 - x =$